Task 3

1) optimization and gradient descent

optimization is the process of finding the best weights that minimize error

Gradient descent is an optimization algorithm which is used to train machine learning models and neural networks. It updates model parameters by gradient.

2) What is the difference between overfitting and underfitting?

Overfitting happens when a model learns too much from the training data, including details that don't matter (like noise or outliers).

Underfitting is the opposite of overfitting. It happens when a model is too simple to capture what's going on in the data.

Overfitting	Underfitting
Model is too complex	Model is too simple
Learns noise instead of patterns	Cannot capture the data pattern
Excellent training performance, poor test performance	Poor performance on training and test data

3) precision and recall and the trade of between them

Precision is a metric that measures how often a machine learning model correctly predicts the positive class.

Precision = TP / (TP + FP)

Recall is a metric that measures how often a machine learning model correctly identifies positive instances (true positives) from all the actual positive samples in the dataset.

Recall = TP / (TP + FN)

- Increasing **precision** decreases false positives
- Increasing **recall** decreases false negatives

4) feature selection

Feature selection is the process of choosing the most relevant features for a model.

Benefits

- Reduces overfitting
- Improve model performance
- Reduces training time
- Lower compute costs

We can use methods like Correlation, Chi-square, Forward selection, Backward elimination to select features

5) What do you know about Ensemble learning?

Ensemble learning combines multiple models to improve prediction performance.

Advantages

- Reduces variance and bias
- Improves accuracy

6) What is Random Forest and how does it work?

A random forest is a machine learning algorithm that uses an "ensemble" of many decision trees to solve classification and regression problems.

How it works

1. Creates many decision trees using random samples of the data
2. Selects random subsets of features at each split
3. Each tree makes a prediction
4. Final prediction is made by majority voting (classification) or averaging (regression)

7) What do you know about backward propagation?

Backpropagation is an algorithm used to train neural networks.

It differs from the traditional feedforward process because it propagates the error backward through the network, allowing earlier layers to update their weights based on their contribution to the final error.